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Interpretable Traffic Sign Recognition

A Dual-Stage Pipeline Integrating YOLOv8 and Prototypical Decision Trees

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1 INTRODUCTION

Localizing and classifying traffic signs

1

Localization

A fine-tuned YOLOv8 detector finds every traffic sign in the driving scene and draws a bounding box around it.

2

Classification

Each detected sign is assigned to one of 200 classes covering the full traffic-sign taxonomy.

3

Interpretability

A prototype decision tree explains each prediction by matching the sign against learned visual prototypes.

Goal: a perception pipeline that accurately detects traffic signs and provides explanations for each classification.

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CONTEXT & MOTIVATION

Accuracy is not enough: safety needs transparency

Safety-critical perception

Traffic-sign recognition drives ADAS and autonomous-driving decisions: a missed stop or speed limit can cause unsafe behaviour.

The black-box problem

Standard CNN detectors output a label and a confidence score but give no human-readable justification for the decision.

Demand for explainability

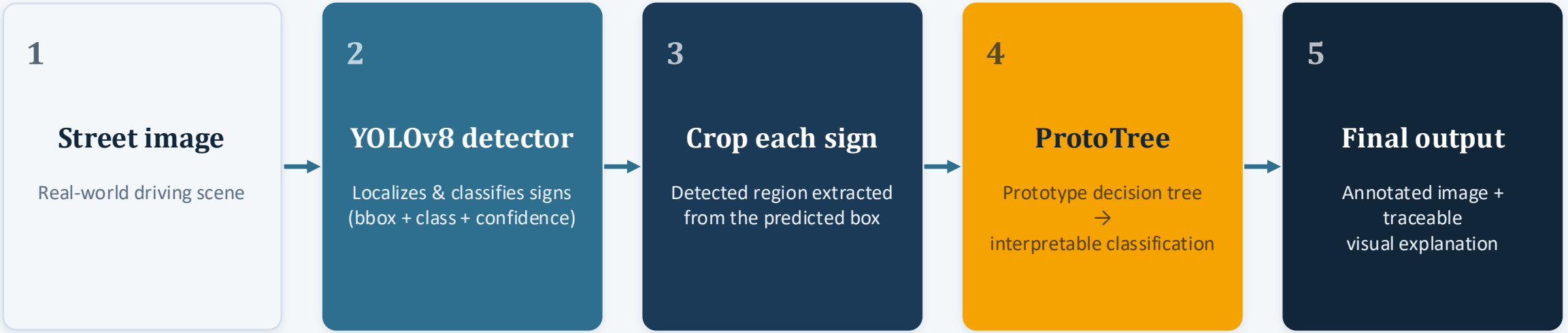
Regulators and users increasingly require interpretable decisions — especially when a system misclassifies or fails.

Our stance: detection accuracy and interpretability are treated as ***complementary, not competing*** objectives.

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PROPOSED APPROACH

A two-stage detect-then-explain pipeline



Each component is optimized and evaluated by criteria appropriate to its role — and ProtoTree explains YOLOv8's decision rather than replacing it.

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DATASET

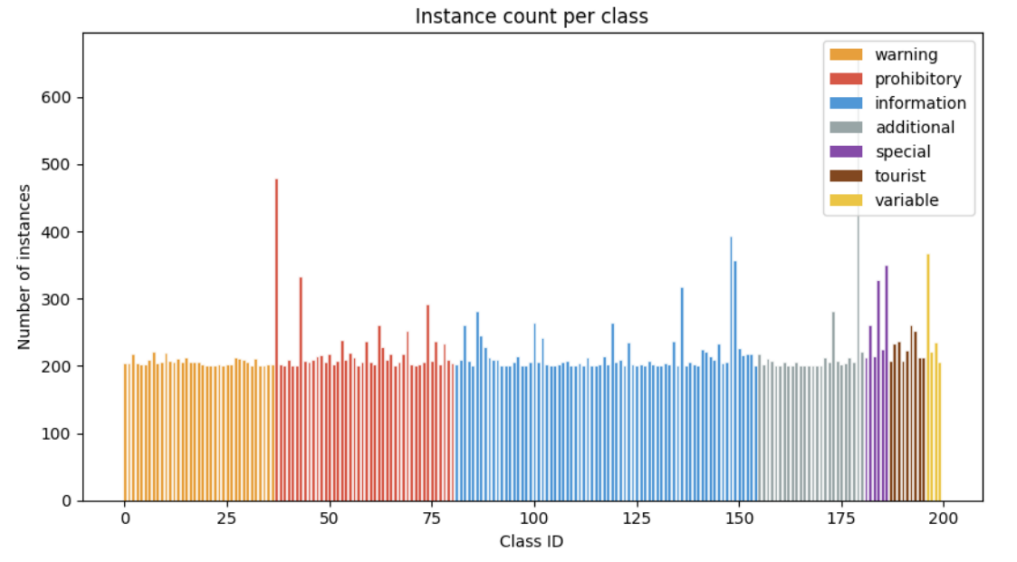
DFG Traffic Sign Dataset — large-scale & balanced

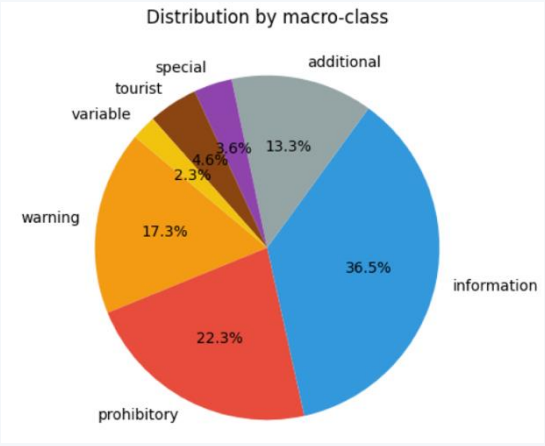
14,029
annotated images

44,091
sign instances

200
sign classes

7
macro-classes





Slovenian road-network signs, distributed approximately uniformly across the 200 classes — **mitigating the long-tailed bias common in traffic-sign benchmarks.**

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METHODOLOGY

Robust splitting & a staged model-selection strategy

Data partitioning

- **Multi-label stratified split**
keeps all 200 classes proportionally represented in every partition.
- **Held-out test set — 15%**
2,085 images / 6,635 instances, never seen during training or selection.
- **5-fold cross-validation**
on the remaining 85%
- **This partitioning is shared at image level by YOLO and ProtoTree**

Phase 1 — Variant screening

5 YOLOv8 variants (n→x) · Fold 0 (50 epochs)

▸ *pick variant → YOLOv8m*

Phase 2 — Hyperparameter search

2 learning rates × 2 freeze strategies · Fold 0 (50 epochs)

▸ *pick config → lr 1e-3, no freeze*

Phase 3 — Cross-validation

Final config × all 5 folds · 80 epochs

▸ *robust performance estimate*

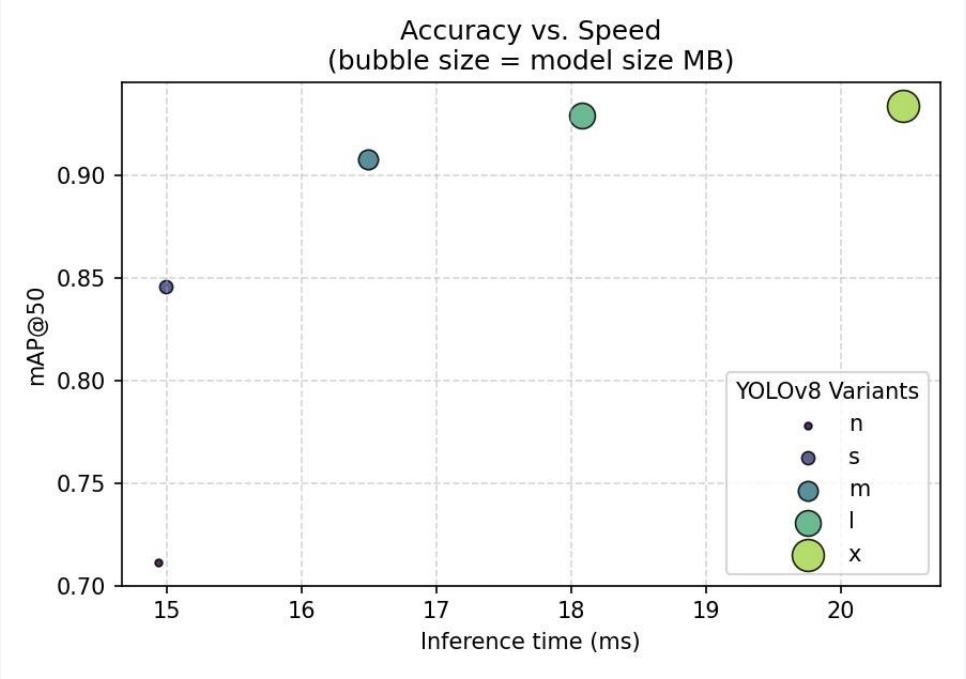
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RESULTS · PHASE 1

Variant comparison: the accuracy–speed trade-off

Variant	mAP@50	mAP@50-95	Prec.	Rec.	Lat. (ms)
n	0.711	0.651	0.701	0.677	14.94
s	0.846	0.787	0.832	0.793	14.99
m	0.908	0.858	0.896	0.858	16.49
l	0.930	0.887	0.923	0.884	18.08
x	0.934	0.893	0.928	0.892	20.46

YOLOv8m selected. Nano→Medium gains +0.197 mAP@50 at negligible latency cost; Medium→XLarge adds only +0.026 for +24% latency and 2× model size.



Curve flattens from the Medium variant onward — diminishing returns.

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RESULTS · PHASE 2

Hyperparameter grid search on YOLOv8m

lr0	Freeze	mAP@50	mAP@50-95	Recall
4.9e-5	—	0.908	0.858	0.858
4.9e-5	10 layers	0.841	0.789	0.786
1e-3	**—**	0.929	0.884	0.883
1e-3	10 layers	0.872	0.818	0.815

Highlighted row = configuration carried into Phase 3.

lr0 = 1×10^{-3}, no frozen layers

Best configuration — highest mAP@50 of 0.929 on Fold 0.

Full fine-tuning wins

Despite the small per-class sample size, unfreezing the whole pretrained backbone consistently outperforms freezing.

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RESULTS · PHASE 3

Cross-validation & held-out test performance

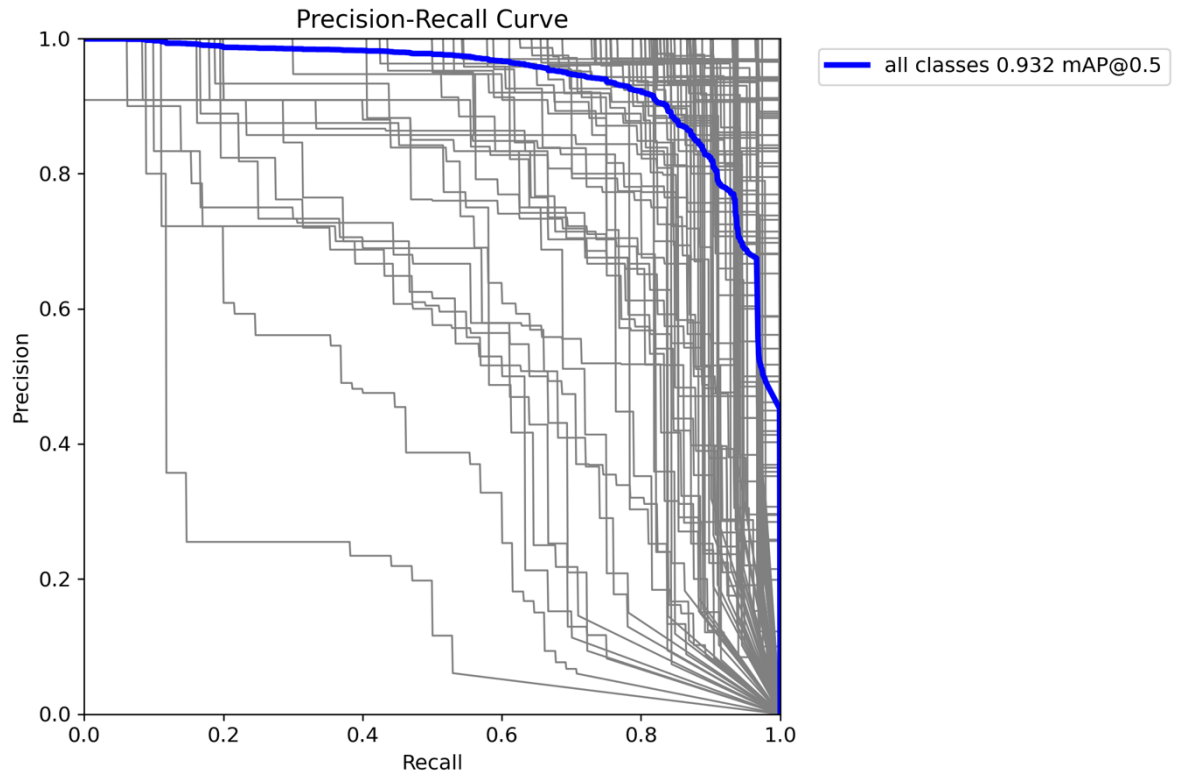
Metric	Mean ± Std. (5-fold)
mAP@50	0.9318 ± 0.0028
mAP@50-95	0.8877 ± 0.0035
Precision	0.9294 ± 0.0051
Recall	0.8910 ± 0.0054

Std. of just 0.0028 on mAP@50 → stable across folds

HELD-OUT TEST SET

mAP@50 0.932 · mAP@50-95 0.892

Precision 0.933 · Recall 0.887



Precision–recall curve on the test set (all classes, 0.932 mAP@0.5).

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INTERPRETABLE CLASSIFICATION

ProtoTree: a traceable tree of visual prototypes

ResNet-50 backbone

ImageNet-pretrained feature extractor

256-d prototype layer

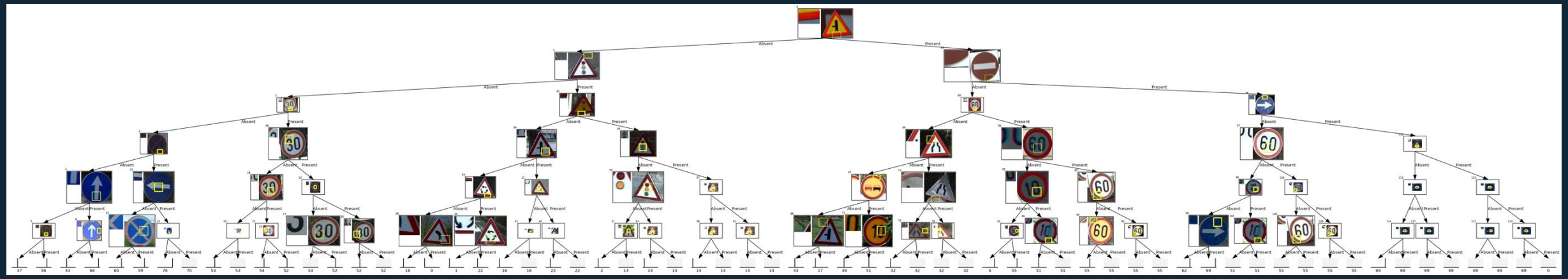
learned visual prototypes ('this looks like that')

Depth 6 · up to 64 leaves

compact tree from the 25-class selection

Prune + project

every node grounded in a real training patch



Pruned & projected ProtoTree — each internal node shows its visual prototype; routing follows the prototype's presence/absence in the crop.

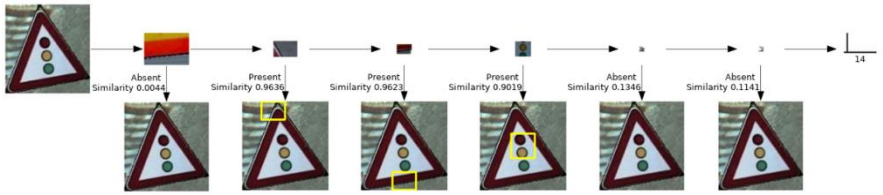
High accuracy with fully traceable decisions

High and uniform per-class performance.
The weakest class still reaches 0.73 recall — a few categories remain hard to disambiguate even on clean crops.

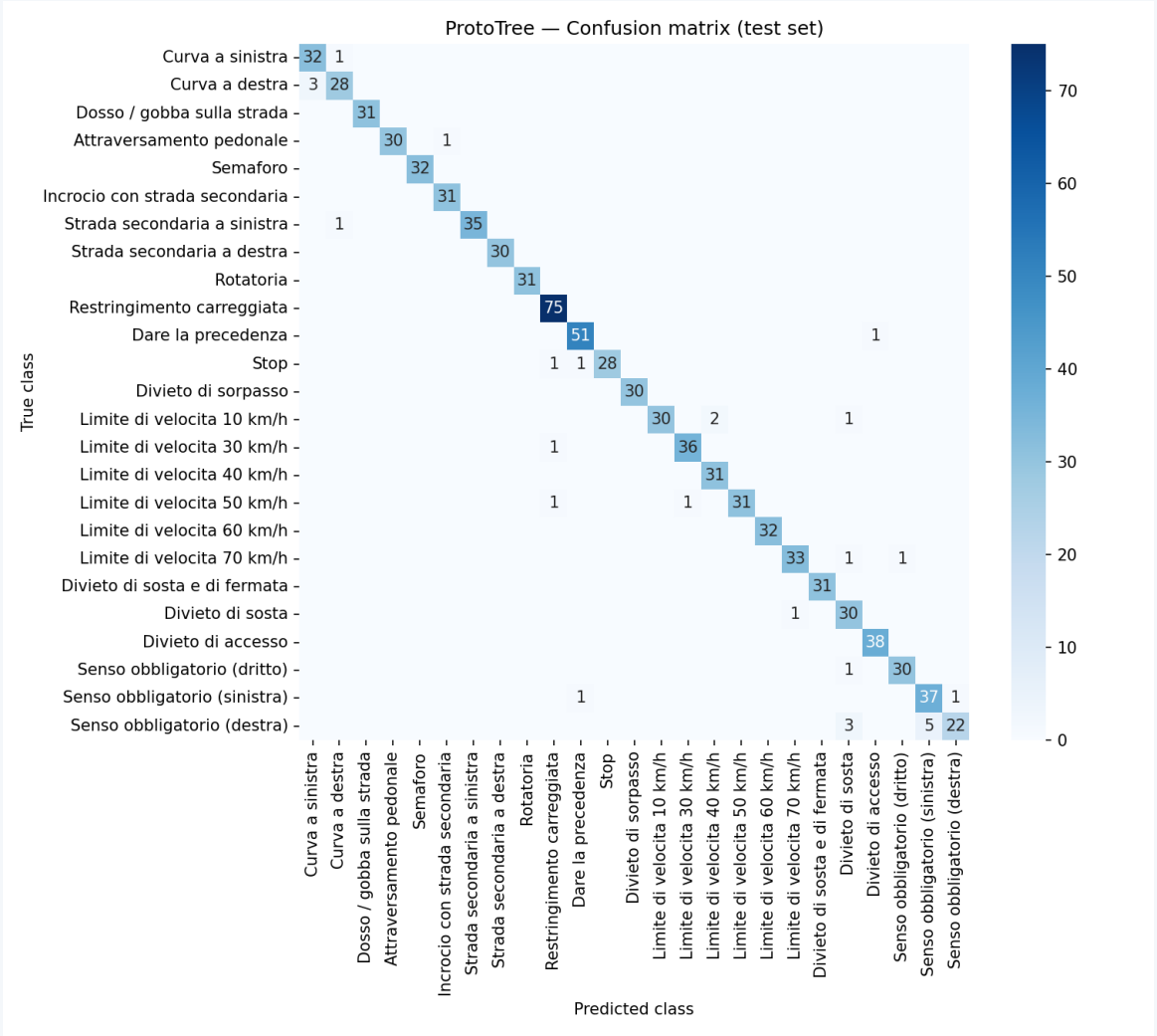
97.5% test accuracy

0.97 precision & recall

874 ground-truth test crops



Local explanation: a root-to-leaf path of prototype matches yields the prediction.



Confusion matrix (test crops)

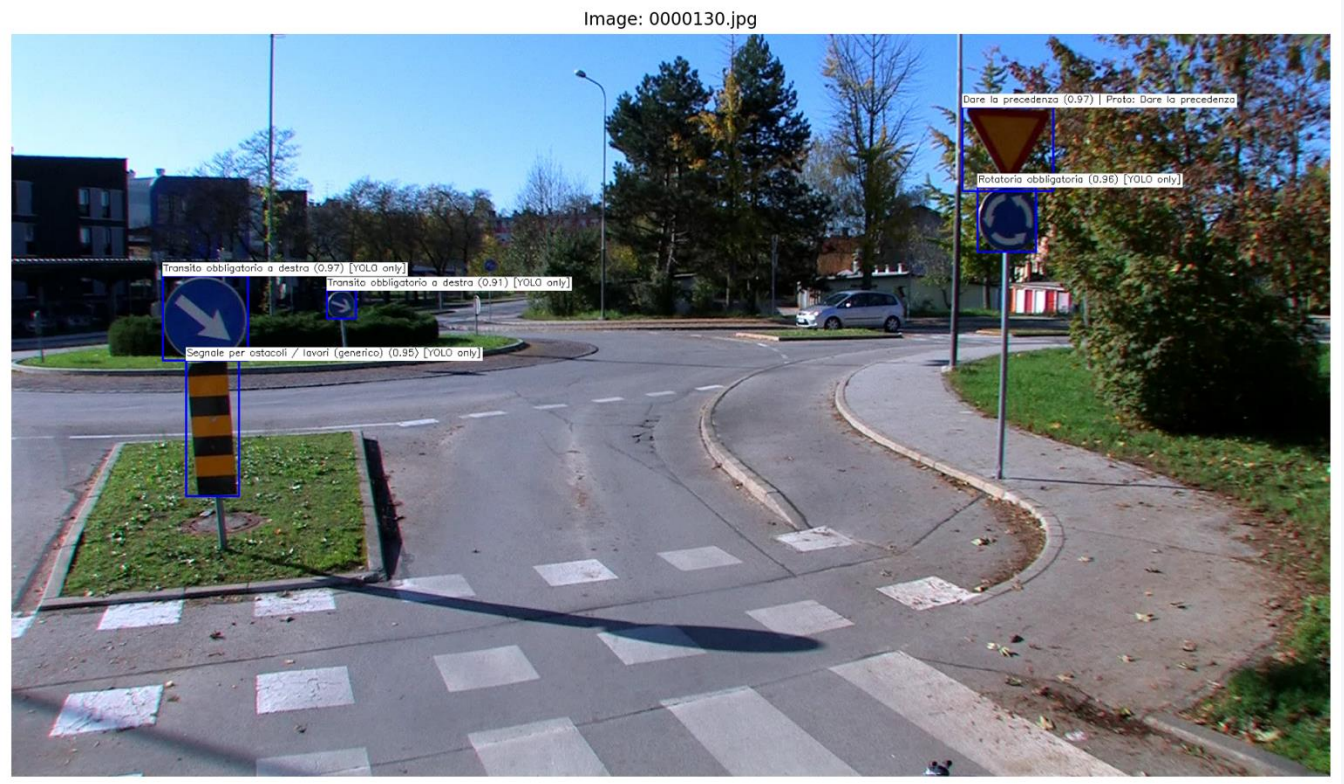
Does the explanation track the detector's decision?

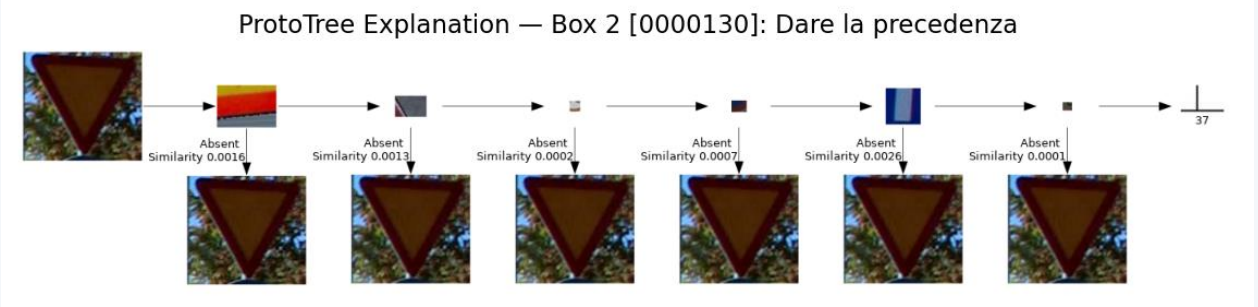
93.2%

agreement between ProtoTree and YOLOv8 on the detector's own predicted crops (785 / 842)

Faithful explanatory proxy.
ProtoTree is trained only on clean ground-truth crops; yet stays consistent with the detector even on imperfect, predicted boxes.

Stage 1 — YOLOv8 localizes the signs in the scene.





Stage 2 — ProtoTree's prototype path explains one localized sign.

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DISCUSSION

Limitations & directions for future work

Limitations

1. **25-class explanatory scope** ProtoTree covers a road-safety subset; full coverage needs a deeper tree or an ensemble of topic-specific trees.

2. **Bounded by the detector** False positives or loose boxes feed corrupted crops downstream — pipeline consistency drops from localization, not classification.

3. **Single-fold early selection** Phases 1–2 used one split for efficiency; full CV from the start would be even more unbiased.

4. **Limited hyperparameter sweep** Search focused on macro-architecture; batch size, finer LRs and ProtoTree structure left unexplored.

Future work

- Hierarchical prototype trees organized by macro-class for broader, still-inspectable coverage.
- Confidence filtering between stages to shield ProtoTree from poorly localized crops.
- Full cross-validation from the very first selection phase.
- Exhaustive grid search over finer hyperparameters.

Video Demo with YOLO



CONCLUSIONS

Detection and interpretability, achieved together

1. **State-of-the-art detection** — YOLOv8m reaches 0.932 mAP@50 on the held-out test set, with stable cross-validated performance.

2. **Transparent by design** — A compact ProtoTree adds 97.5% accurate, visually-grounded explanations — without touching detector accuracy.

3. **Faithful on real outputs** — 93.2% agreement shows the explanation tracks the detector even on its own imperfect crops.